IE643 Lecture 7

Announcement:

Quiz-1. Date: 1st September 2026 (Tuesday) Venue: In class.

Syllabus: Contents till Perceptron mistake bound analysis. Open scribes/notes. More instructions to be posted in moodle.

Recap:

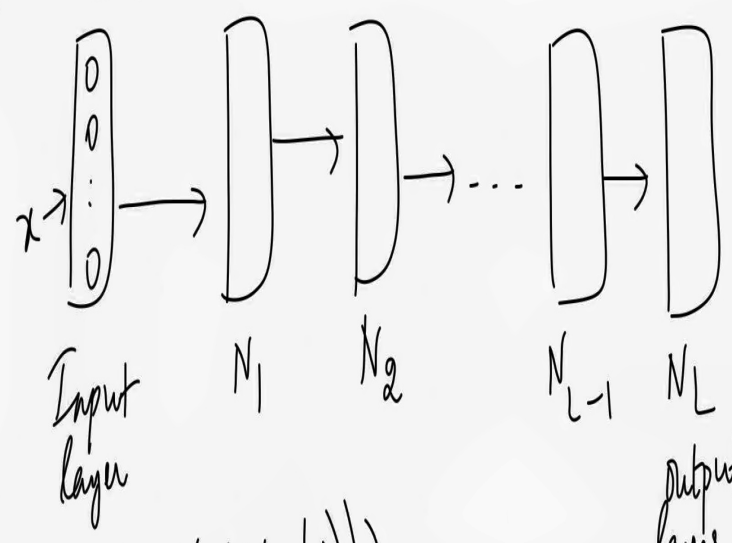


Figure 1: A chain of $N_1 \to N_2 \to \cdots \to N_{L-1} \to N_L$ nodes (layers) — input layer ($x \in \mathbb{R}^d$) feeding into N_1, then each layer feeding into the next until output layer.

$$\hat{y} = a^L = \phi(W^L; \phi(W^{L-1}\phi(W^2\phi(W^1 x))))$$

$$e(y, \hat{y}) \to \text{loss fn to compare true and predicted labels}$$

output layer.

Supervised learning

$$D = \{(x^i, y^i)\}_{i=1}^N$$

$$x \ni x^i \in \mathbb{R}^d$$

$$y \ni y^i \in \{+1, -1\}$$

$$h : X \to Y$$

MLP weights

train a multilayered perceptron to find h_{MLP}

$$\min E = \sum_{i=1}^{N} e(y^i, \hat{y}^i)$$

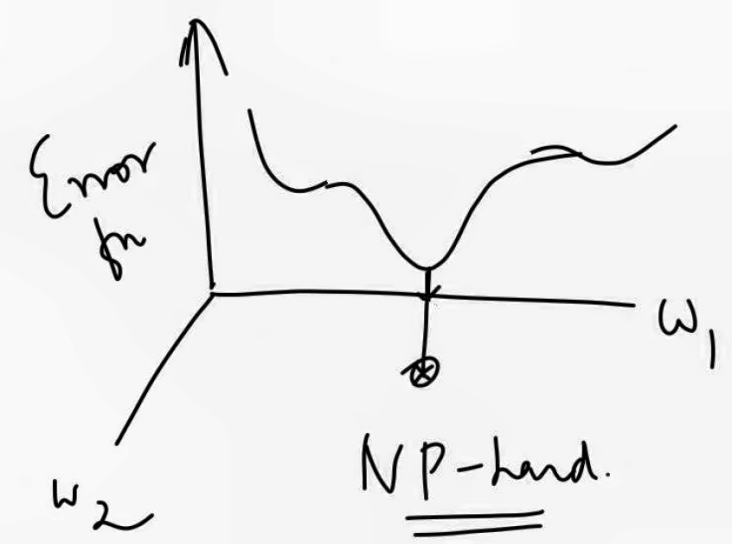


Figure 2: A 3D error surface plot with axes w_1, w_2, showing a bowl-shaped curve with a local minimum marked by an “x”.

NP-hard

local minimum

minimization is over weights of MLP

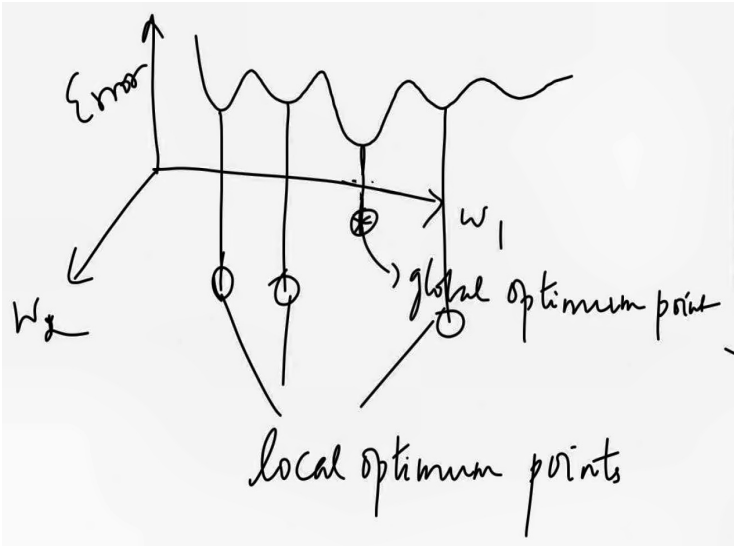


Figure 3: A wavy error curve with several local minimum "bowl" basins labeled "local optimum points", one of which is marked "global optimum point".

Iterative procedure to find a local minimum point

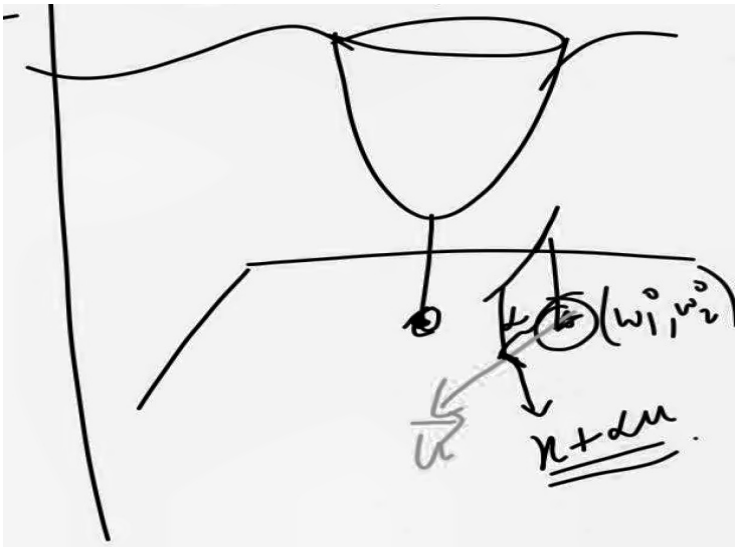


Figure 4: A bowl-shaped contour plot with a point at w^* and a small step Δw drawn from it, labeled "$w + \Delta w$".

Directional derivative:

For a fn. $f : \mathbb{R}^d \to \mathbb{R}$, the directional derivative along some **non-zero** direction $u \in \mathbb{R}^d$ at a point x is defined as

$$f'(x; u) = \lim_{\alpha \downarrow 0} \frac{f(x + \alpha u) - f(x)}{\alpha}$$

Gradient of a fn at a point x

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix}$$

$$f'(x;u) = \langle \nabla f(x), u \rangle$$

$$u = -\nabla f(x) = -\|\nabla f(x)\|^2$$

If $\nabla f(x) \neq$ zero vector:

$$f'(x;u) = \langle \nabla f(x), -\nabla f(x) \rangle$$

$$f'(x; -\nabla f(x)) < 0$$

Descent direction:

A non-zero direction $u \in \mathbb{R}^d$ is called a descent direction of f at x if:

$$f'(x;u) < 0$$

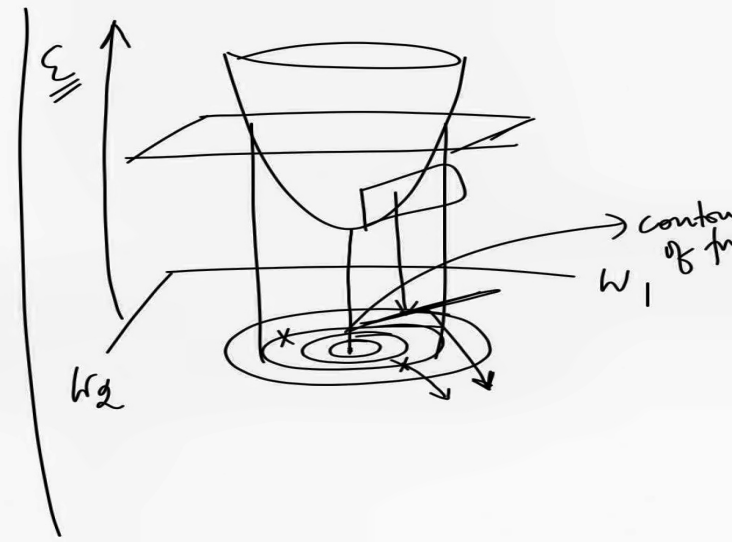


Figure 5: A 3D surface with axes η_1, η_2, η_3 (height axis) — a bowl-shaped contour plot with level curves labeled w_1, w_2, etc., illustrating the gradient direction toward the minimum.

Negative gradient direction $-\nabla f(x)$

step-length / learning rate.

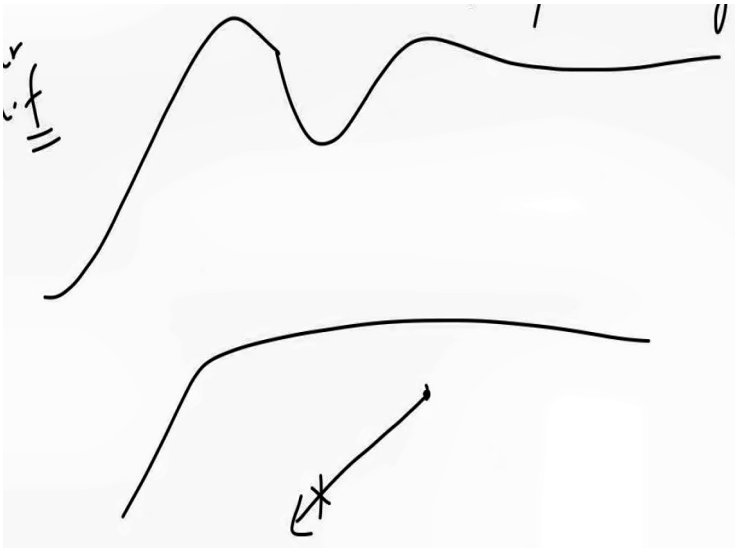


Figure 6: A wavy curve with an arrow pointing left from a marked point, illustrating the negative gradient direction.

$$f'(x,u) = \lim_{\alpha \downarrow 0} \frac{f(x+\alpha u) - f(x)}{\alpha}$$

Property:

If u (a non-zero vector) is a descent direction of fn $f : \mathbb{R}^d \to \mathbb{R}$ at x, then:

there exists some $\epsilon \geq 0$ such that for every $\alpha \in [0, \epsilon]$, $\lim_{\alpha \downarrow 0} \frac{f(x+\alpha u) - f(x)}{\alpha} < 0$

$$f(x+\alpha u) < f(x) \quad \Rightarrow \quad f'(x;u) < 0$$

$$\min_w E = \sum_{i=1}^{N} e(y^i, \hat{y}^i) \qquad \nabla E = \sum_{i=1}^{N} \nabla e(y^i, \hat{y}^i) \rightarrow \text{sample-based gradient}$$

full gradient → very large → overfit

- w^0
- for $k = 0, 1, 2, ...$
  - find descent dirn using $-\nabla E(w^k)$ (10^{-8})
  - find a suitable step length α_k
  - $w^{k+1} \leftarrow w^k + \alpha_k(-\nabla E(w^k))$
  - zero-gradient check to stop this process

ϵ_k: fixed to be small value close to 0; $10^{-5}, 10^{-4}, ...$

α_k: step-length finding procedures

Stochastic Gradient Descent

$$D = \{(x^i, y^i)\}_{i=1}^N$$

$\rightarrow w^0$

$\rightarrow$ for $k = 0, 1, 2, ...$ $\hookrightarrow$ Compute a **stochastic gradient** $\rightarrow$ randomly choose $j \in \{1, 2, 3 \dots N\}$ $\rightarrow$ Compute $\nabla e(y^j, \hat{y}^j)$

$$w^{k+1} \leftarrow w^k + \eta_k \nabla e(y^j, \hat{y}^j)$$

Stochastic approx. 51: convergence to the local opt/saddle pt.

might not lead to descent

Mini-batch SGD

$$D = \{(x^i, y^i)\}_{i=1}^N$$

$\rightarrow w^0$

$\rightarrow$ for $k = 0, 1, 2, ...$

$\hookrightarrow$ Sample mini-batch of samples:

$$B_k \subseteq \{1, 2, \dots, N\}$$

for every $j \in B_k$: compute $\nabla e(y^j, \hat{y}^j)$

$\hookrightarrow w^{k+1} \leftarrow w^k + \eta_k \left(- \sum_{j \in B_k} \nabla e(y^j, \hat{y}^j) \right)$

convergence in asymptotic sense

Adam

SGD with momentum

$$m_k = \alpha \cdot \text{grad}_k + (1 - \alpha) m_{k-1}$$

generalize perf

$$v^k = \alpha(\text{mini-batch gradient}) + (1 - \alpha) \cdot \text{past gradients}$$

Adam:

$$w^{k+1} \leftarrow w^k - \eta_k$$

AMSGrad:

$$\left(\frac{\text{running avg of gradients}}{\sqrt{\text{running avg of sq. grad.}} + \epsilon} \right)$$

$$m_k = \beta_1 \cdot \text{grad}_k + (1 - \beta_1) m_{k-1}$$

$$v_k = \beta_2 \cdot (\text{grad}_k^2) + (1-\beta) v_{k-1}$$

$$w^{k+1} \leftarrow w^k + \eta_k \left(-\frac{m_k}{\sqrt{v_k} + \epsilon} \right)$$

Sample-based gradient

$$\nabla e(y^j, \hat{y}^j)$$

Computation of sample-based gradient

$$\hat{y}^j = \phi(W^2 \phi(W^1 x^j)) \cdots$$

$$D_e(y^j, \hat{y}^j) \to w_1, w_2, w_3, \ldots, w^L$$

auto-differentiation

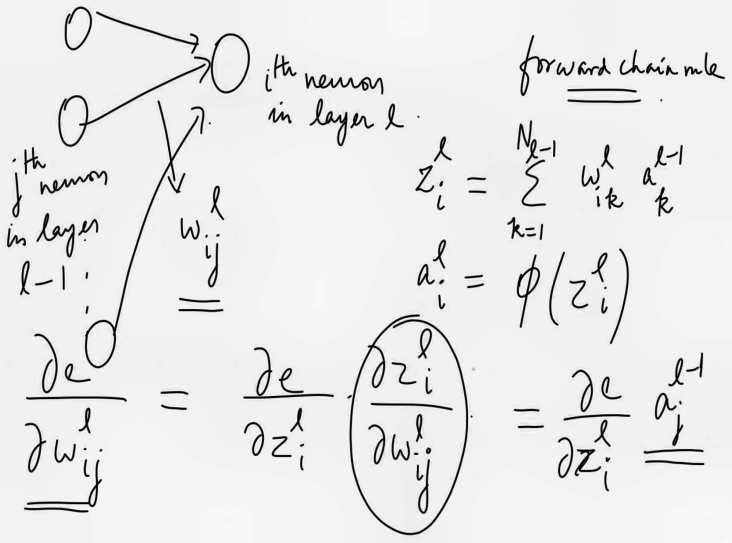


Figure 7: A feedforward network — j^{th} neuron in layer $l-1$ connects via weight w_{ij}^l to i^{th} neuron in layer l.

i^{th} neuron in layer l

j^{th} neuron in layer $l-1$

$$\frac{\partial e}{\partial w_{ij}^l} = \frac{\partial e}{\partial z_i^l} \cdot \boxed{\frac{\partial z_i^l}{\partial w_{ij}^l}}$$

forward chain rule

$$z_i^l = \sum_{k=1}^{N_{l-1}} w_{ik}^l a_k^{l-1}$$

$$a_i^l = \phi(z_i^l)$$

$$= \frac{\partial e}{\partial z_i^l} a_j^{l-1}$$

$$\frac{\partial e}{\partial z_i^l} = \left(\frac{\partial a_i^l}{\partial z_i^l}\right) \cdot \phi'(z_i^l) = \frac{\partial e}{\partial a_i^l} \cdot \phi'(z_i^l)$$

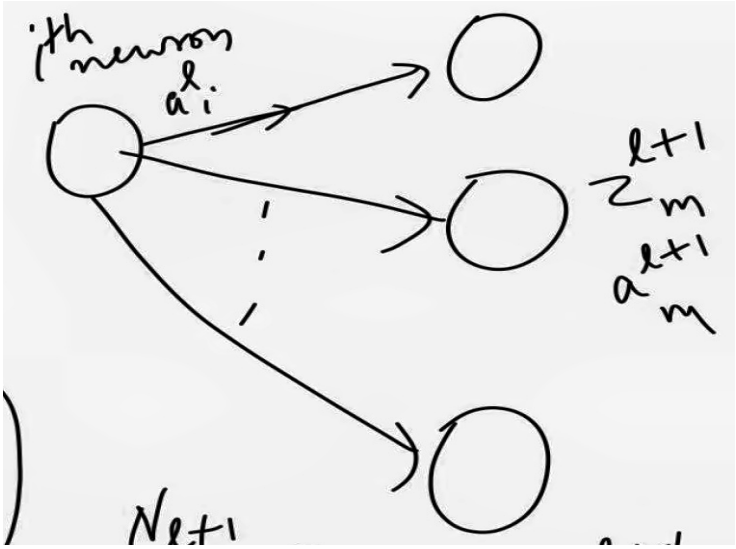


Figure 8: A neuron a_i^l (layer l) feeding into N_{l+1} neurons of layer $l+1$, labeled z_m^{l+1}, a_m^{l+1} $(m = 1, \ldots, N_{l+1})$.

$$\frac{\partial e}{\partial a_i^l} = \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial z_m^{l+1}} \cdot \left(\frac{\partial z_m^{l+1}}{\partial a_i^l}\right) = \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial z_m^{l+1}} \cdot w_{mi}^{l+1}$$

$$z_m^{l+1} = \sum_{k=1}^{N_l} w_{mk}^{l+1} a_k^l$$

$$\frac{\partial e}{\partial a_i^l} = \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial z_m^{l+1}} \cdot w_{mi}^{l+1}$$

$$= \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial a_m^{l+1}} \cdot \frac{\partial a_m^{l+1}}{\partial z_m^{l+1}} \cdot w_{mi}^{l+1}$$

$$= \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial a_m^{l+1}} \phi'(z_m^{l+1}) \cdot w_{mi}^{l+1}$$

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